

Question 1

Marks

- a) Evaluate, correct to four significant figures, $\sqrt[4]{\frac{4^5 - 5^4}{17 + 7^3}}$. 1
- b) Solve $\frac{|5 - x|}{7} \geq 3$ Graph your solution on a number line. 3
- c) Solve $x(5 - x) = 8 - x$. 2
- d) Find a primitive of $5 - \frac{1}{e^x}$. 2
- e) Simplify $\frac{x^2 + x + 1}{5x^3 - 5} + \frac{7}{x - 1}$ 2
- f) Five 78 seater coaches, all full, can bus the entire population of a certain school to Homebush in 3 trips each. If a different bus company was engaged for the return journey, using seven small buses, each seating 28, how many trips would each bus need to travel? 2

Question 2 Start a new page

Marks

- a) The graph of a parabolic function crosses the x -axis at the origin and at $x = 4$. If the minimum value of the function is -12 , determine the parabolic function. 3
- b) A right isosceles triangle has one vertex at the origin O and another at the point $A(1,3)$. The base of the triangle has equation $x = 2y$.
- (i) Show that the third vertex B has coordinates $(4,2)$. 3
- (ii) Find the length of OA . 1
- (iii) Find the area of the triangle. 2
- (iv) Find the length of OB . 1
- (v) Find the perpendicular height of the triangle. *from A to OB.* 2

Question 3 Start a new page

Marks

- a) Evaluate $\int_0^2 \frac{x}{5-x^2} dx$. 2
- b) Jesse loves to play tennis. One day Jesse hopes be a champion tennis player. Jesse's coach says that her skills improve by about 5% with every competition match. How many matches will Jesse need to play so that her game skills are at lease twice as good as they are right now? 2
- c) Differentiate with respect to x :
- (i) $\sin(3x^2+4x)$. 2
- (ii) $\frac{\ln 5x}{e^{7x}}$. 2
- d) A particle moves in a straight line. At time t seconds its distance x metres from a fixed point O in the line is given by $x = 3 \cos \pi t - 3$.
- (i) Sketch the graph of x as a function of t for $0 \leq t \leq 2$. 1
- (ii) Show that the time when the particle first comes to rest is $t = 1$. 1
- (iii) In two or more sentences, describe the motion of the particle during the first second. 2

Question 4 Start a new page

Marks

- a) Without sketching the function, determine the set

of x values for which $y = \frac{6}{x^2 - 1} - 3$ is defined

and write down any x and y intercepts.

3

- b) State the range and domain of the functions

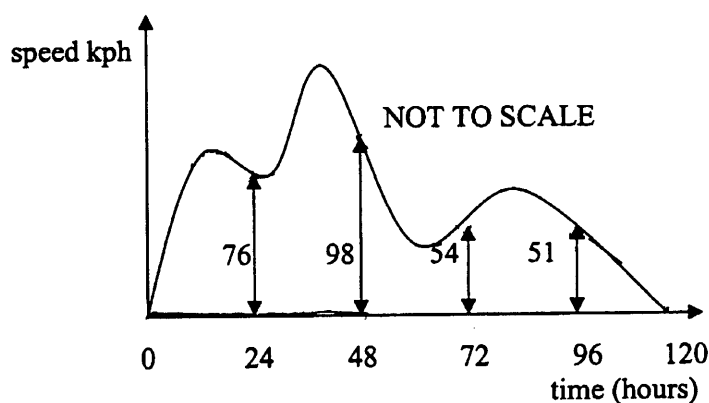
(i) $g(x) = e^{3\sin 2x}$.

2

(ii) $f(x) = \log_e(e^{3\sin 2x})$.

2

- c)



The curve graphed above represents the speed of a hot air balloon during a 5 day adventure. The area under the speed graph represents the total distance travelled during the flight.

- (i) Use the trapezoidal rule to obtain an approximate value for the total distance travelled.

3

- (ii) State whether you believe that the actual distance travelled by the balloon is more or less than the answer obtained in part (i) above. Give reasons for your answer. You may include a diagram or sketch.

2

Question 5 Start a new page

Marks

- a) Show that zero is the least integer value of k for which the quadratic equation $(k + 1)x^2 - x + 1 = 0$ has no real roots. 2
- b) Consider the function $y = x^4 - 4x^3$.
- (i) On a neat set of coordinate axes sketch the function showing points of inflection, intercepts on axes and turning points. 2
- (ii) On the same set of axes sketch the gradient function showing its points of inflection, turning points and intercepts. 2
- c) (i) Sketch the curves $y = 1 + \frac{1}{x}$ and $y = \sin(2x)$, for $0 \leq x \leq \pi$.
[Use the same coordinate axes for both] 2
- (ii) Shade the area represented by $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 1 + \frac{1}{x} - \sin(2x) dx$. 1
- (iii) Find the value of the shaded area as a simplified exact value. 3

Question 6 Start a new page.

Marks

a) The 3rd term of an arithmetic sequence is -6 and the 23rd term is 9.

(i) Show that the 24th term is 9.75 2

(ii) Sum the terms from the 3rd to the 23rd inclusive. 2

(iii) Show that 99 is a term of the sequence. 1

b) The function $f(x) = 5xe^{\frac{-x}{2}} + 3$

has first derivative $f'(x) = -\frac{5x}{2}e^{\frac{-x}{2}} + 5e^{\frac{-x}{2}}$

and second derivative $f''(x) = \frac{5x}{4}e^{\frac{-x}{2}} - 5e^{\frac{-x}{2}}$.

(i) Find the coordinates of the stationary point. 1

(ii) Find the values of x for which $f(x)$ is increasing. 1

(iii) Find the values of x for which $f(x)$ is decreasing. 1

(iv) Find the values of x for which $y=f(x)$ is concave up. 1

(v) Find the values of x for which $y=f(x)$ is concave down. 1

(vi) Find the y intercept and $f(6)$. 1

(vii) Determine the absolute minimum of $f(x)$ for $0 \leq x \leq 6$ 1

Question 7 Start a new page

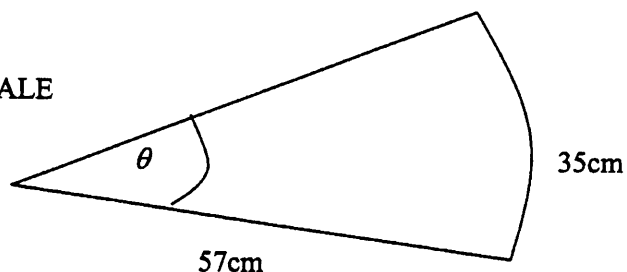
Marks

- a) The region enclosed between the curves $y = -x^2$ and $x = y^2$ is rotated about the x axis. Find the volume of the solid of revolution.

3

b)

NOT TO SCALE



Find the size of angle θ , shown in the diagram above.

Give your answer correct to the nearest minute.

2

- c) The time elapsed during the motion of a particle is

given by $t = \frac{5}{(x+3)} - 1$.

- (i) Explain why the particle can never move to a position where $x = 3$
- (ii) Show that the displacement of the particle, in terms of t , is given by $x = \frac{5}{1+t} - 3$
- (iii) Does the displacement approach a limiting position? Explain your answer.
- (iv) Find an expression for velocity in terms of time.
- (v) Does the particle ever appear to stop moving? Explain your answer.

1

1

2

1

2

Question 8 Start a new page

Marks

- a) The n^{th} term of the series $0 + \frac{7}{5} - \frac{21}{50} + \dots$ is given by the

formula $A\left(\frac{1}{5}\right)^n + B\left(\frac{-1}{2}\right)^n$.

- (i) By writing $T_1 = 0$, find the values of the constants A and B
and hence calculate the 4th term of the series.

3

- (ii) Show that the sum of the series to n terms

is given by $\frac{5}{2}\left(1 - \left(\frac{1}{5}\right)^n\right) + \frac{-4}{3}\left(1 - \left(\frac{-1}{2}\right)^n\right)$

2

- (iii) Find the sum to infinity.

2

- b) During the normal operation of a petrol driven engine, the
volume V litres of petrol left in the tank reduces at a rate

$$\frac{dV}{dt} = -3e^{0.4t} \text{ where } t \text{ is measured in minutes since the engine}$$

was switched on and the tank was full (100 Litres).

- (i) At what rate is the petrol used, initially?

1

- (ii) Use integration to show that volume remaining can

be expressed as $V = \frac{-30}{4}e^{0.4t} + 107.5$

2

- (iii) How long can the machine operate until the tank is

only half full? Answer correct to the nearest second.

2

Question 9 Start a new Page

Marks

- a) The number of bubbles appearing on the surface of a glass of lemonade decreases over time until eventually the lemonade is described as 'flat'. Marcus observed the lemonade when it was freshly poured. Initially, Marcus counted 72 bubbles. 16 seconds later there were only 24 bubbles. Assume that the number of bubbles satisfies the equation $N = N_0 e^{-kt}$ where N_0 and k are constants and t is measured in seconds.
- (i) Find values of k and N_0 and predict when there will be only 2 bubbles observed. 3
- (ii) How many seconds will pass from there being 2 bubbles until there is only 1 bubble observed. 2
- b) Consider a **straight line** with equation $3y = mx + 6$, $m > 0$, and a **curve** with equation $y = \log_e(x+1)$.
- (i) By substituting $m=2$, sketch $3y = 2x + 6$ and $y = \log_e(x+1)$ together on the same diagram. 2
- (ii) Show that the vertical distance from the line to the curve is given by the expression $\frac{2x}{3} + 2 - \log_e(x+1)$. 1
- (iii) Now for the more general case where $3y = mx + 6$ and $y = \log_e(x+1)$, the **vertical** distance from the straight line to the curve is given by $\frac{mx}{3} + 2 - \log_e(x+1)$. Show that the shortest vertical distance is given the expression $3 - \frac{m}{3} - \log_e(\frac{3}{m})$ 3
- (iv) Find the value of m so that the shortest vertical distance coincides with the y -axis. 1

Question 10 Start a new page

Marks

a) Solve the equation $\tan \pi x = \frac{1}{\sqrt{3}}$ for $0 \leq x \leq 2$. 2

b) Assume that a plant leaf will grow under suitable conditions until nutrients are in short supply, then the leaf will stop growing and it will maintain its size. A simple mathematical model for the growth of a certain type of plant leaf involves the following **split function**

$$\left. \begin{array}{l} W = W_0 e^{\mu t} \text{ for } 0 \leq t < t_f \\ W = W_f \text{ for } t \geq t_f \end{array} \right\}$$

where

W represents dry weight of the leaf at time t

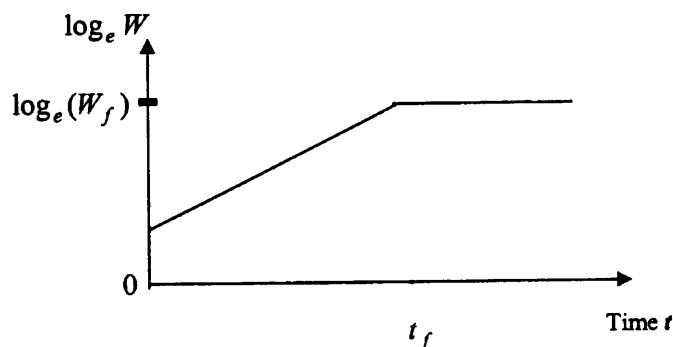
μ represents 'growth rate', $\mu > 0$

t_f represents time when the leaf stops growing

W_f represents dry weight of the leaf at time t_f

W_0 represents initial dry weight of the leaf, $W_0 > 1$

- (i) Show by differentiation that the equation $W = W_0 e^{\mu t}$ may be written as $\frac{dW}{dt} = \mu W$ for $0 \leq t < t_f$ 1
- (ii) Show that the **split function** is never decreasing 1
- (iii) Sketch the **split function** on a set of axes with W on the vertical axis and t on the horizontal axis. 2
- (iv) Study the graph of $\log_e W$ against time t , shown below.



For the sloping line segment above; show that the gradient is μ ; determine the intercept on the vertical axis and write its equation as function of time. 3

- (v) Now, under experimental conditions using the same plant, additional nutrients are provided causing leaves to continue growing until they reach twice the usual size. Show that the additional time taken for the leaf to grow is given by $\frac{\log_e 2}{\mu}$ 3

END OF PAPER.

Roshana

0 a) $\sqrt[4]{\frac{4^5 - 5^{-4}}{17 + 73}}$

= 1.02604... (1/2)

≈ 1.026 (1/2) for 4

①

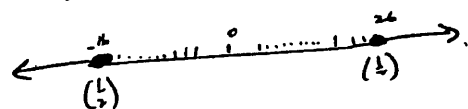
b) $\frac{|5-x|}{7} \geq 3$
 $5-x \geq 21$

∴ $5-x \geq 21$ or $5-x \leq -21$

$-x \geq 16$ or $-x \leq -26$
 $x \leq -16$ (1/2) $x \geq 26$ (1/2)

(No ± in for direction incorrect of sign)

(No ± mark if not filled in for 1 of the answers)



c) $x(5-x) = 8-x$
 $5x - x^2 = 8-x$
 $-x^2 + 6x - 8 = 0$
 $-(x^2 - 6x + 8) = 0$
 $-(x-4)(x-2) = 0$
 $x = 2 \text{ or } 4$ (1/2, 1/2)

$5x - x^2 = 8-x$
 $x^2 - 6x + 8 = 0$
 $(x-4)(x-2) = 0$
 $x = 2 \text{ or } 4$
 $\frac{1}{2}$

②

d) $\int 5x e^{\frac{1}{x}} dx$
 $= \int 5 e^{-x} dx$ (2)
 $= 5x + e^{-x} + C$
 $= 5x + \frac{1}{e^x} + C$

or $\int 5 - \frac{1}{e^x} dx$ (1m)
 $= 5x + \dots + C$

e) $\frac{x^2+x+1}{5x^2-5} + \frac{7}{x-1}$
 $\frac{x^2+x+1}{5(x-1)(x^2+x-1)} + \frac{7}{x-1}$
 $= \frac{x^2+x+1 + 7(x^2+x-1)}{5(x-1)(x^2+x-1)}$

or 4

②

f) 1st trip: people = $5 \times 78 \times 3 = 1170$ (1)

2nd trip: $1170 \div (7 \times 28)$
 $(\frac{1}{2} \text{ for } 5.6 \text{ hours}) = 5.693 \dots$
 $= 6 \text{ TRIPS}$ (1)

e) $\frac{(x^2+x+1)(x-1)}{(5x^2-5)(x-1)} + \frac{7(5x^2-5)}{(5x^2-5)(x-1)}$ (1/2)

= $\frac{x^3 - x^2 + x^2 - x + x - 1 + 35x^2 - 35}{(5x^2-5)(x-1)}$

= $\frac{36x^2 - 36}{(5x^2-5)(x-1)}$

= $\frac{36(x^2-1)}{(5x^2-5)(x-1)}$

= $\frac{36(x-1)(x^2+x-1)}{5(x^2-1)(x-1)}$

= $\frac{36(x^2+x-1)}{5(x^2-1)}$

= $\frac{36(x^2+x-1)}{5(x-1)(x^2+x-1)}$ (1/2)

= $\frac{36}{5(x-1)}$ (1/2)

or

or $\frac{36(x^2-1)}{5(x^2-1)(x-1)}$ (1/2)

$\frac{36}{5(x-1)}$ (1/2)

Question 2

Part a

Method 1 $y = ax(x-4)$ (1)
 Sub in $(2, -12)$, $a=3$ (1)
 $\therefore$ eqn is $y = 3x(x-4)$ (1)

Method 2 $y = a(x-2)^2 - 12$ No a (2) only
 Sub $(0,0)$, $a=3$
 $\therefore y = 3(x-2)^2 - 12$ (3)

Method 3 $(x-2)^2 = 4a(y+12)$ (2)
 or $(x-2)^2 = 4(y+12)$ (No a)
 Sub $(1,0)$, $a = \frac{1}{12}$
 $y = 3(x-2)^2 - 12$
 $(x-2)^2 = 4(\frac{1}{12})(y+12)$
 $(x-2)^2 = \frac{1}{3}(y+12)$ (3)

2 (b)

(i) Prove $(4,2)$ lies on the line $x=2y$ (1)

Either prove $OA \perp AB$
 or $OA = AB$ (2)

(ii) $OA = \sqrt{3^2 + 1^2} = \sqrt{10}$ (1)

(iii) Area $= \frac{1}{2} \cdot OA \cdot OB = \frac{1}{2} \sqrt{10} \times \sqrt{10} = 5 \text{ s.u.}$
 or Area $= \frac{1}{2} \cdot OB \times h = \frac{1}{2} (2\sqrt{5})(\sqrt{5}) = 5 \text{ s.u.}$ (2)

(iv) $OB = \sqrt{4^2 + 2^2} = \sqrt{20}$ (1)

(v) Method 1
 Area $= \frac{1}{2} \cdot OB \times h$
 $5 = \frac{1}{2} \times 2\sqrt{5} \times h$
 $h = \sqrt{5}$

Method 2. $x-2y=0$ (Eqn of OB)

Perp. dist from $A(1,3)$ to OB
 $= \frac{|1 \times 1 - 2 \times 3 + 0|}{\sqrt{1^2 + 2^2}}$
 $= \frac{|1-6|}{\sqrt{5}} = \frac{|-5|}{\sqrt{5}} = \sqrt{5}$

Method 3

Find mid-pt of OB: $M(2,1)$
 Distance $AM = \sqrt{5}$

(2)

QUESTION 3.

$$\begin{aligned} \text{a)} \quad & \int_0^2 \frac{x}{5-x^2} dx \\ &= -\frac{1}{2} \int_0^2 \frac{-2x}{5-x^2} dx \\ &= \left[-\frac{1}{2} \ln(5-x^2) \right]_0^2 \\ &= -\frac{1}{2} (\ln 1 - \ln 5) \\ &= \frac{\ln 5}{2} \end{aligned}$$

b). G - game skills now
 $G \times 1.05 = \text{after 1 match.}$
 $G \times 1.05 \times 1.05 = \text{after 2 matches}$
 $G \times 1.05^n = 2G$
 $1.05^n = 2$
 $\ln 1.05^n = \ln 2$
 $n = \frac{\ln 2}{\ln 1.05}$
 $= 14.2$
 $\therefore 15 \text{ matches.}$

or

T_1	T_2	T_n
$1.05g$	$1.05(1.05)g$	$2g$

$$T_n = ar^{n-1}$$

$$2g = 1.05g r^{n-1}$$

$$2 = 1.05 (1.05)^{n-1}$$

$$\ln\left(\frac{2}{1.05}\right) = \ln 1.05^{n-1}$$

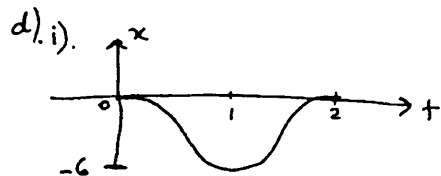
$$13.2 = n-1$$

$$\therefore n = 14.2$$

$\therefore$ need 15 matches.

c). i) $y = \sin(3x^2 + \pi x)$
 $y' = 6x + \pi \cos(3x^2 + \pi x)$

ii). $y = \frac{\ln 5x}{e^{7x}}$
 $y' = \frac{e^{7x} \left(\frac{1}{x} \right) - \ln 5x (7e^{7x})}{(e^{7x})^2}$
 $= \frac{e^{7x} \left(\frac{1}{x} - 7 \ln 5x \right)}{(e^{7x})^2}$
 $= \frac{1}{e^{7x}} \left(\frac{1}{x} - 7 \ln 5x \right)$



ii). $\ddot{x} = -3\pi \sin \pi t$
 $= 0 \text{ for } t = 0, 1, 2, \dots$
 $\therefore$ 1st comes to rest at $t = 1$

iii). Particle starts from rest
 & moves in a negative direction.
 It slows down & comes to rest when $t = 1$ at a negative distance of 6 units from its starting position.

Question 4

(a) all real x except $x \neq \pm 1$ ①

x intercept $y = 0$ $\frac{6}{x^2-1} = 3$
 $6 = 3x^2 - 3$ $\left. \begin{array}{l} -1 \text{ mistake here} \\ 3x^2 = 9 \\ x^2 = 3 \\ x = \pm\sqrt{3} \end{array} \right\}$

y intercept $x = 0$ $y = -6 - 3 = -9$ 3

(b) (i) $g(x) = e^{3 \sin x}$ $D: \text{all real } x$ 1

$-1 \leq \sin x \leq 1$

$-3 \leq 3 \sin x \leq 3$ 2

Range: $\frac{1}{e^3} \leq y \leq e^3$

(ii) $f(x) = \log_e e^{3 \sin x}$

$= 3 \sin x$ $\frac{1}{2}$

$D: \text{all real } x$ 2

$R: -3 \leq f(x) \leq 3$ 1

(c) $d = \int v dt$

$-\frac{1}{2}$ each mistake

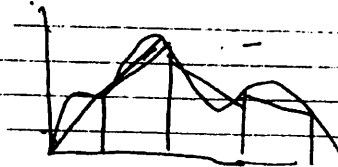
(i) $\int v dt = \frac{24}{2} [0 + 2 \times 76 + 2 \times 98 + 2 \times 54 + 2 \times 51 + 0]$

$= 12 \times 558$

$= 6696 \text{ km}$

[0 - Simpson's Rule] 3

(iii)



More, as curve concave down as trapezoids are under curve.

(2 marks if explains) 2

Q5

$$\begin{aligned} a) \Delta &= (-1)^2 - 4(k+1)(1) \\ &= 1 - 4k - 4 \\ &= -4k - 3 \end{aligned}$$

No real roots $\Rightarrow \Delta < 0 \quad \frac{1}{2}$

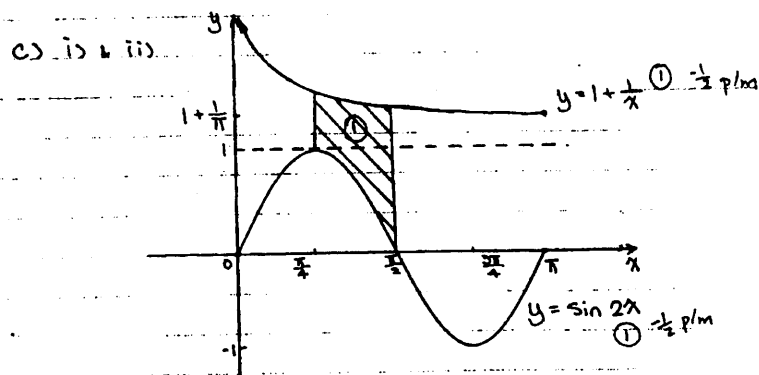
$$-4k - 3 < 0$$

$$-4k < 3$$

$$k > -\frac{3}{4}$$

(2)

Since $k > -\frac{3}{4}$, zero is the least integer value of k .



$$\begin{aligned} \text{iii) } \int_{\pi/4}^{\pi/2} \left(1 + \frac{1}{x} - \sin 2x \right) dx &= \left[x + \ln x + \frac{1}{2} \cos 2x \right]_{\pi/4}^{\pi/2} \\ &= \left(\frac{\pi}{2} + \ln \frac{\pi}{2} + \frac{1}{2} \cos \pi \right) - \left(\frac{\pi}{4} + \ln \frac{\pi}{4} + \frac{1}{2} \cos \frac{\pi}{2} \right) \\ &= \frac{\pi}{4} + \ln \frac{\pi}{2} - \ln \frac{\pi}{4} - \frac{1}{2} \\ &= \ln 2 + \frac{\pi}{4} - \frac{1}{2} \end{aligned}$$

(3)

Q5

$$\begin{aligned} b) y &= x^4 - 4x^3 \\ &= x^3(x - 4) \end{aligned}$$

x -intercept = 0 & 4.

$$\begin{aligned} y' &= 4x^3 - 12x^2 \\ 4x^2(x - 3) &= 0 \end{aligned}$$

$$x = 0 \text{ OR } x = 3$$

$$y = 0 \quad y = -27$$

$$y'' = 12x^2 - 24x$$

$$\text{When } x = 3, y'' = 12 \times 3^2 - 24 \times 3 > 0$$

$\therefore (3, -27)$ is a min. turning pt.

$$\text{When } x = 0, y'' = 0$$

$$x \quad -1 \quad 0 \quad 1$$

$$y'' \quad + \quad 0 \quad -$$

$\therefore (0, 0)$ is a horizontal pt. of inflexion

$$y'' = 12x^2 - 24x$$

$$12x(x - 2) = 0$$

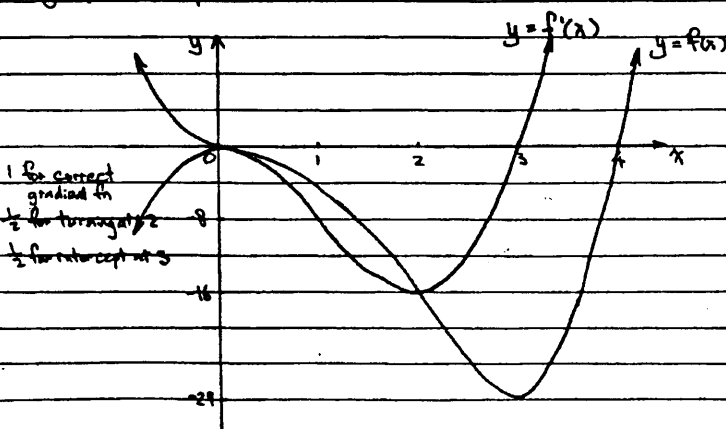
$$x = 0 \text{ OR } x = 2$$

$$y = -16$$

$$x \quad 1 \quad 2 \quad 3$$

$$y'' \quad - \quad 0 \quad +$$

$\therefore (2, -16)$ is a pt. of inflexion



Q6

a) $T_8 = -6$
 $T_{28} = 9$

$$T_8 = a + 7d = -6$$

$$T_{28} = a + 27d = 9$$

$$\therefore 20d = 15$$

$$d = 3/4$$

(1/2)

$$\therefore a + 2 \times 3/4 = -6$$

$$a = -7\frac{1}{2}$$

(1/2)

i) $T_{24} = a + 23d$
 $= -7\frac{1}{2} + 23 \times 3/4$
 $= 9.75$

(1)

ii) $S_{21} = \frac{21}{2}(-6 + 9)$

$$= 31\frac{1}{2}$$

(2)

iii) $99 = a + (n-1)d$
 $99 = -7\frac{1}{2} + (n-1) \times 3/4$
 $143 = n$

(1/3)

$\therefore$ since n is an integer, 99 is a term of the sequence

(1/2)

b) i) stat pts occur when $f'(x) = 0$

$$-\frac{5x}{2}e^{-\frac{x}{2}} + 5e^{-\frac{x}{2}} = 0$$

$$e^{-\frac{x}{2}} \left(-\frac{5}{2}x + 5\right) = 0$$

$$e^{-\frac{x}{2}} \neq 0 \quad -\frac{5}{2}x + 5 = 0$$

$$5x = 10$$

$$x = 2$$

(1/2)

when $x=2$, $f(x) = 5 \times 2 \times e^{-1} + 3$
 $= \frac{10}{e} + 3$

$$(2, \frac{10}{e} + 3)$$

(1/3)

ii) $f(x)$ is increasing when $f'(x) > 0$

$$e^{-\frac{x}{2}} \left(-\frac{5}{2}x + 5\right) > 0$$

$$-\frac{5}{2}x + 5 > 0$$

$$x < 2 \quad (1)$$

iii) $f(x)$ is decreasing when $f'(x) < 0$

$$\therefore x > 2 \quad (1)$$

iv) concave up when $f''(x) > 0$

$$5e^{-\frac{x}{2}} \left(\frac{1}{2}x - 1\right) > 0$$

$$\frac{1}{2}x - 1 > 0$$

$$x > 4 \quad (1)$$

v) concave down when $f''(x) < 0$

$$\therefore x < 4 \quad (1)$$

vi) y-intercept

$$f(0) = 3$$

(1/2)

$$f(6) = \frac{30}{e^3} + 3$$

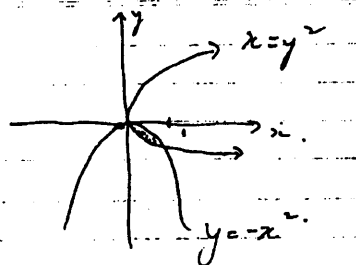
$$\approx 4.49 \quad (2 \text{ dec. pl})$$

(1/2)

vii) $\therefore$ absolute minimum for $0 \leq x \leq 6$ is 3.

(1)

Q7. a)



$$y^2 = x^4$$

$$\therefore x = x^4$$

$$x(x^3 - 1) = 0$$

$$x^3 = 1 \Rightarrow x = 1$$

$$x = 1 \text{ or } 0$$

$$V = \pi \int y^2 dx$$

$$V = \pi \int_0^1 (x^2 - x^4) dx$$

$$= \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1$$

$$= \pi \left[\frac{1}{3} - \frac{1}{5} - 0 \right]$$

$$= \frac{2\pi}{15}$$

b)

$$35 = 57.0^\circ$$

$$\theta = \frac{35}{57.0} \text{ radians}$$

$$= \frac{35}{57.0} \times \frac{180}{\pi} \text{ degrees}$$

$$= 35^\circ 11' \text{ nearest minute}$$

$$c) t = \frac{5}{x+3} - 1$$

$$(i) x = 3 \quad t = \frac{5}{6} - 1$$

$$t = -\frac{1}{6}$$

$$\text{but } t > 0 \therefore x \neq 3$$

$$(ii) t+1 = \frac{5}{x+3}$$

$$x+3 = \frac{5}{t+1}$$

$$x = \frac{5}{t+1} - 3$$

$$(iii) t \rightarrow \infty \quad x \rightarrow -3$$

$$\therefore \text{approaches horizontal asymptote of } x = -3$$

$$(iv) v = \dot{x} = \frac{-5}{(t+1)^2}$$

$$(v) \dot{x} = \frac{-5}{(t+1)^2} \neq 0$$

$$\text{but as } t \rightarrow \infty \quad \dot{x} \rightarrow 0$$

Q 8

$$T_1 = 0 = A\left(\frac{1}{5}\right) + B\left(-\frac{1}{2}\right)$$

$$0 = \frac{1}{5}A - \frac{1}{2}B$$

$$0 = 2A - 5B \quad (1)$$

$$T_2 = \frac{7}{5} = \frac{1}{25}A + \frac{1}{4}B$$

$$140 = 4A + 25B \quad (2)$$

$$2 \times (1) \quad 140 = 35B$$

$$4 = B$$

$$\text{and } 2A = 5 \times 4 = 20$$

$$A = 10$$

$$T_4 = 10 \times \left(\frac{1}{5}\right)^4 + 4 \times \left(\frac{1}{2}\right)^4$$

$$= \frac{2}{125} + \frac{1}{4}$$

$$= \frac{133}{500} \text{ or } 0.266$$

(ii) Series is $10\left(\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \dots + \frac{1}{5^n}\right) + 4\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \left(\frac{1}{2}\right)^n\right)$

$$S_n = 10\left(\frac{1}{5} \left(\frac{1 - \left(\frac{1}{5}\right)^n}{1 - \frac{1}{5}}\right)\right) + 4\left(\frac{1}{2} \left(\frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}}\right)\right)$$

$$= 10\left(\frac{1}{4} \left(1 - \left(\frac{1}{5}\right)^n\right)\right) + 4 \times \left(\frac{1}{3} \left(1 - \left(\frac{1}{2}\right)^n\right)\right)$$

$$= \frac{10}{4} \left(1 - \left(\frac{1}{5}\right)^n\right) - \frac{4}{3} \left(1 - \left(\frac{1}{2}\right)^n\right)$$

$$= \frac{5}{2} \left(1 - \left(\frac{1}{5}\right)^n\right) - \frac{4}{3} \left(1 - \left(\frac{1}{2}\right)^n\right)$$

$$S_{\infty} = \frac{5}{2} (1 - 0) - \frac{4}{3} (1 - 0) \quad \left| \begin{array}{l} \text{OR} \\ S_{\infty} = 10\left(\frac{1}{5}\right) + 4\left(\frac{1}{2}\right) \\ = 10 \times \frac{1}{5} + 4 \times \frac{1}{2} \\ = \frac{1}{6} \end{array} \right. \quad 2$$

b) (i) $\frac{dV}{dt} = -3e^{0.4t}$

when $t = 0$

$$\frac{dV}{dt} = -3 \text{ l/min}$$

petrol reducing at rate of 3 l/min

(ii) $V = -3 \int e^{0.4t} dt$

$$= -\frac{3 \times 5}{2} e^{0.4t} + c \quad \text{with or without } c$$

when $t = 0 \quad V = 100$

$$100 = -\frac{15}{2} e^0 + c$$

$$\therefore c = 107.5$$

$$V = -\frac{30}{4} e^{0.4t} + 107.5$$

(iii) $50 = -\frac{30}{4} e^{0.4t} + 107.5$

$$57.5 \times \frac{4}{30} = e^{0.4t}$$

$$7.6 = e^{0.4t}$$

$$e^{0.4t} = 7\frac{2}{3}$$

$$0.4t \ln e = \ln 7\frac{2}{3}$$

$$\frac{t}{0.4} = \frac{1}{0.4} \ln \left(7\frac{2}{3}\right)$$

$$t = 5.09220418 \text{ mins}$$

$$t = 5 \text{ mins } 6 \text{ secs}$$

Q9 $N_0 = 72$
 $N_{16} = 24$
 i) $24 = 72 e^{-16k}$
 $\frac{1}{3} = e^{-16k}$
 $3 = e^{16k}$
 $k = \frac{1}{16} \ln 3$
 $2 = 72 e^{-\ln 3^{\frac{1}{16}} \cdot t}$
 $\frac{1}{36} = e^{-\ln 3^{\frac{1}{16}} \cdot t}$
 $36 = e^{\ln 3^{\frac{1}{16}} \cdot t}$
 $\frac{\ln 36}{\ln 3^{\frac{1}{16}}} = t$
 $t = 52.189$ sec

P9 ctd

part 9) ctd.

$$1 = 72 e^{-\ln 3^{\frac{1}{16}} t}$$

$$72 = e^{\ln 3^{\frac{1}{16}} t}$$

$$\ln 72 = t$$

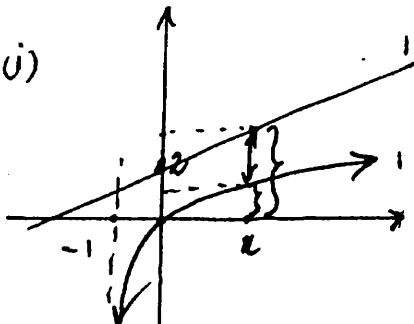
$$\ln 3^{\frac{1}{16}}$$

$$t = 62.28 \dots$$

$\times 62$ seconds

$\therefore 10$ more seconds

b/ i)



2)

i.e. by subtracting ordinates.

$$(ii) y = \frac{2x}{3} + 2 - \ln(x+1)$$

$$(iii) D = \frac{mx}{3} + 2 - \ln(x+1)$$

$$\frac{dD}{dx} = \frac{m}{3} - \frac{1}{x+1}$$

$$= 0 \text{ for } x+1 = \frac{3}{m}$$

(iv)

$$x = \left(\frac{3}{m} - 1\right)$$

$$\frac{d^2D}{dx^2} = \frac{1}{(x+1)^2}$$

> 0 for all x ($x \neq -1$)

$\therefore$ minima at

$$x = \frac{3}{m} - 1$$

$$= 1 - \frac{m}{3} + 2 - \ln \frac{3}{m}$$

$$= -3 - \frac{m}{3} - \ln \frac{3}{m}$$

(iv) from $x = \frac{3}{m} - 1$

$m = 3$ for $x = 0$
 to coincide with the y-axis.

Q 10

Q10 $\tan^{-1} x = \frac{1}{\sqrt{3}} \quad 0 \leq x \leq 2$

$\pi x = \frac{\pi}{6} \quad \frac{7\pi}{6} \quad 0 \leq \pi x \leq 2\pi$

$= \frac{1}{6}, \frac{7}{6}$

b) $W = W_0 e^{\mu t}$

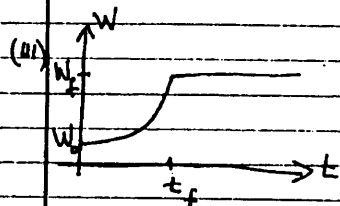
$\frac{dW}{dt} = \mu W_0 e^{\mu t}$

$= \mu W \quad \text{for } 0 \leq t < t_f$

(i) for $0 \leq t < t_f \quad \frac{dW}{dt} = \mu W > 0$ as $\mu > 0$ and $W > 0$

for $t \geq t_f \quad \frac{dW}{dt} = 0$

Hence $\frac{dW}{dt} \neq 0$ for all values of t i.e. it is never decreasing



5 things
-1 for anything missing

iv $\log W = \log W_0 e^{\mu t}$
 $= \log W_0 + \mu t$
 $= \mu t + \log W_0$

gradient = μ
OR

$\frac{d \log W}{dt} = \frac{W'}{W} = \frac{\mu W}{W} = \mu$

OR

gradient = $\frac{\log W_f - \log W_0}{t_f}$ now $W_f = W_0 e^{\mu t_f}$

$= \frac{\log \left(\frac{W_f}{W_0} \right)}{t_f}$

$= \frac{\log \left(\frac{W_0 e^{\mu t_f}}{W_0} \right)}{t_f}$

$= \frac{\mu t_f}{t_f}$
 $= \mu$

y intercept = $\log W_0$

(v) Let additional time be t i.e. time to get from $W_f \rightarrow 2W_f$

$2W_f = W_f e^{\mu t}$

$2 = e^{\mu t}$

$\ln 2 = \mu t$

$\frac{\ln 2}{\mu} = t$

3